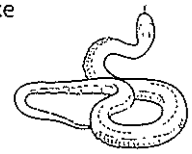
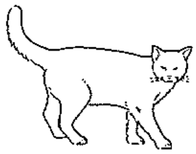
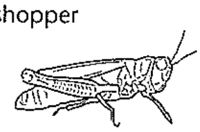
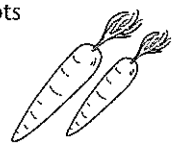
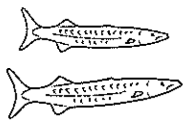
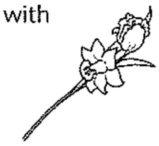
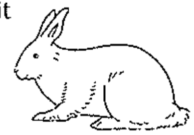
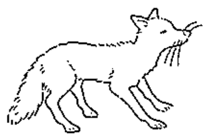
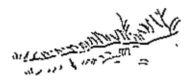
Food chains and webs

17a

Food chains show how energy moves through a community by looking at what is eaten by what.

snake 	small fish 	mouse 	grass seeds 
kookaburra 	cat 	grasshopper 	hummingbird 
carrots 	large fish 	algae 	tadpoles 
flowers with nectar 	rabbit 	fox 	grass 

1 Cut out each square containing an organism above and arrange them into four separate food chains. Don't forget to include the arrow (→ = energy goes to, is eaten by). Paste your food chains into your exercise book, labelling each organism as either a producer, first order consumer, second order consumer and so on.

Looking at language

Text can be written in active voice or passive voice.

In active voice the person or thing that is doing something is placed at the beginning of the clause, because that is the main subject of the clause.

For example: *The cat* ate the bird.

In passive voice the person or thing that something is happening to is at the beginning of the clause, because this is what the clause is about. And whoever is doing the action is placed towards the end of the clause, if it is stated at all.

For example: The bird was eaten by *the cat*. The bird was eaten.

2 Use your food chains from question 1 to complete the table below by writing clauses in active voice and passive voice. One pair of clauses has been done for you.

Active voice	Passive voice
The rabbit eats the grass.	The grass is eaten by the rabbit.
The snake	The mouse
The small fish	
The mouse	
The tadpoles	
The fox	
The grasshopper	
The large fish	

Food chains and webs

3 Below is a table showing different organisms found in Ms Smart's back yard. She watched them all day and recorded her observations in the table below.

Organism	What it feeds on
Grass	Makes its own food (producer)
Mangrove tree ▪ leaves ▪ fruit ▪ flowers ▪ branches	(producer)
Bottle brush ▪ leaves ▪ flowers ▪ branches	(producer)
Bees	Flowers and fruits (herbivore)
Dragonflies	Bees and ladybirds (carnivore)
Aphids	Sugars from leaf veins of mango (herbivore)
Stick insect	Leaves from bottlebrush (______________________)
Insect borer	Mango branches (______________________)
Ladybirds	Aphids (______________________)
Wagtail	Insect borers and stick insects (______________________)
Honeyeater	Flowers and fruits (______________________)
Kookaburra	Stick insects and dragonflies (______________________)
Cat	The birds wagtail and honeyeater (______________________)
Flying fox	Mango fruits (______________________)
Possum	Mango fruit and bottlebrush flowers (______________________)

a Complete Ms Smart's table; she forgot to add the labels *carnivore* and *herbivore*.

b Use Ms Smart's information to construct a food web in your exercise book.

4 Ms Smart accidentally sprayed her mango tree and bottlebrush tree with insecticide. As a result all the aphids and insect borers were killed. Explain what could happen to the food web.

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5 a What would happen if Ms Smart's insect population triples? Explain what could happen to the food web straight away.

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b If the insect population had tripled, explain what would happen over the following weeks and months.

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